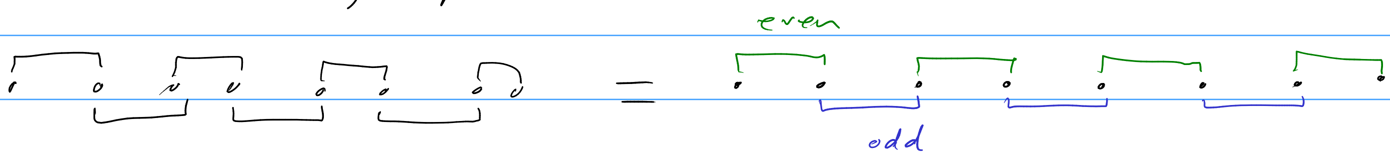
$$h = -it \sum_i \gamma_i \gamma_{i+1} + g \sum_i \gamma_i \gamma_{i+2} \gamma_{i+3} \gamma_{i+4} + \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+4}$$

split hamiltonian into h_{even}, h_{odd} and rewrite majorana operators into even and odd:

$$a_j := \gamma_{2n}$$
$$b_j := \gamma_{2n+1}$$



$$h = h_{even} + h_{odd}$$

$$= -it\left(\sum_j a_j b_j + \sum_j b_j a_{j+1}\right)$$

→ Use Jordan Wigner transformation to map fermionic (majorana) operators to spin operators (Pauli matrices)

$$a_j = K_j \sigma_j^x$$
$$b_j = K_j \sigma_j^y$$
$$a_j b_j = i \sigma_j^z$$
$$K_j := \prod_{l=1}^{j-1} \sigma_l^z$$

$$-it\left(\sum_j i\sigma_j^z + K_j \sigma_j^y \underline{K_{j+1}} \sigma_{j+1}^x\right)$$

$$= -it\left(\sum_j \left(i\sigma_j^z + K_j \sigma_j^y K_j \sigma_j^z \sigma_{j+1}^x\right)\right)$$

$$-it\left(\sum_j i\sigma_j^z + \cancel{K_j^2} \underbrace{\sigma_j^y \sigma_j^z}_{=i\sigma_j^x} \sigma_{j+1}^x\right)$$

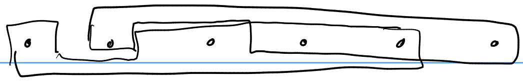
$$\sigma_i \sigma_j = i \epsilon_{ijk} \sigma_k$$

$$= -it\left(\sum_j i\sigma_j^z + i\sigma_j^x \sigma_{j+1}^x\right)$$

$$= \sum_j -\overset{h}{\cancel{t}}\sigma_j^z - \overset{J}{\cancel{t}}\sigma_j^x \sigma_{j+1}^x = \sum_j -J \sigma_j^x \sigma_{j+1}^x - h\sigma_j^z \quad \leftarrow \text{as expected.}$$

let $t = J = h$

$$g\sum_i \gamma_i\gamma_{i+2}\gamma_{i+3}\gamma_{i+4} = g\sum_j a_j a_{j+1} b_{j+1} a_{j+2} + b_j b_{j+1} a_{j+2} b_{j+2}$$

 $= a_j\; b_j\; a_{j+1}\; b_{j+1}\; a_{j+2}\; b_{j+2}$

$$a_j a_{j+1} b_{j+1} a_{j+2} =$$

$$K_j\sigma_j^x \cdot i\sigma_j^z \cdot K_{j+2}\sigma_{j+2}^x = \cancel{K_j}\sigma_j^x\, i\sigma_{j+1}^z \cdot \cancel{K_j}\sigma_j^z\sigma_{j+1}^z\sigma_{j+2}^x$$

$$= \sigma_j^x\, i\sigma_{j+1}^z\,\sigma_j^z\,\sigma_{j+1}^z\,\sigma_{j+2}^x$$

$$i\sigma_j^x\sigma_j^z\,\underbrace{\cancel{\sigma_{j+1}^z\sigma_{j+1}^z}}_{1}\,\sigma_{j+2}^x$$

$$i(-i\sigma_j^y)\sigma_{j+2}^x = \sigma_j^y\sigma_{j+2}^x$$

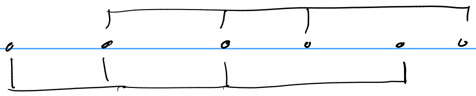
$$b_j b_{j+1} a_{j+2} b_{j+2} = \cancel{K_j}\sigma_j^y \cdot \cancel{K_j}\sigma_j^z\sigma_{j+1}^y \cdot i\sigma_{j+2}^z$$

$$\sigma_j^y\sigma_j^z\sigma_{j+1}^y\, i\sigma_{j+2}^z$$

$$\cancel{i}\,i\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z = -\sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z$$

$$g\sum\gamma_i\gamma_{i+2}\gamma_{i+3}\gamma_{i+4} = g\sum\sigma_j^y\sigma_{j+2}^x - \sigma_j^x\sigma_{j+1}^y\sigma_{j+2}^z$$

$+\gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = \sum_j a_j b_j a_{j+1} a_{j+2} + b_j a_{j+1} b_{j+1} b_{j+2}$

 $= a_j \; b_j \; a_{j+1} \; b_{j+1} \; a_{j+2} \; b_{j+2}$

$$\cancel{a_j b_j} \cdot a_{j+1} \cdot a_{j+2} = i\sigma_j^z \cdot \cancel{K_j} \sigma_j^z \sigma_{j+1}^x \cdot \cancel{K_j} \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^x$$

$$= i \sigma_j^z \cancel{\sigma_j^z \sigma_j^z} \sigma_{j+1}^x \sigma_{j+1}^z \sigma_{j+2}^x$$

$x \; y \; z$

$z \; x \quad i\sigma_{j+1}^y$

$$i\sigma_j^z (-i\sigma_{j+1}^y) \sigma_{j+2}^x = \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x$$

$$b_j a_{j+1} b_{j+1} b_{j+2} = \cancel{K_j} \sigma_j^y \cdot i\sigma_{j+1}^z \cdot \cancel{K_j} \sigma_j^z \sigma_{j+1}^z \sigma_{j+2}^y$$

$$= \sigma_j^y \sigma_j^z \left(i \underbrace{\cancel{\sigma_{j+1}^z \sigma_{j+1}^z}}_{1} \right) \sigma_{j+2}^y$$

$$= (i\sigma_j^x)(i)(\sigma_{j+2}^y) = -\sigma_j^x \sigma_{j+2}^y$$

$$g \sum \gamma_i \gamma_{i+1} \gamma_{i+2} \gamma_{i+3} = g \sum \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y$$

→ flipped sign of second term and reversed x, y, z ndex order

↳ makes sense.

$$h = -t \sum_j \left(\sigma_j^z + \sigma_j^x \sigma_{j+1}^x\right) + g \sum \left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z\right) + \left(\sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y\right)$$

$$h = -t \sum_j \left(\sigma_j^z + \sigma_j^x \sigma_{j+1}^x\right) + g \sum_j \left(\sigma_j^y \sigma_{j+2}^x - \sigma_j^x \sigma_{j+2}^y\right) - \left(\sigma_j^x \sigma_{j+1}^y \sigma_{j+2}^z - \sigma_j^z \sigma_{j+1}^y \sigma_{j+2}^x\right)$$